

Introduction

Robots and humanoids have always ruled our imagination. Last month, we covered the FIRA event in Bangalore and were excited to see how robots battled in a game of soccer. Besides, it's exciting to see how a machine behaves when it is endowed with logic to decide, interpret and react to its surroundings. In fact, it's difficult to find someone who is not fascinated by “intelligent” machines. Machines that can walk, talk and see invokes the kid in each one of us. Robots have always attracted hobbyists and professionals alike.

There has always been a dearth of good books that teach you how to build a robot yourself. Pick up any book on robotics, and soon you'll be overwhelmed with the daunting mathematical equations, dreadful diagrams and a feeling that building a robot is too technical and difficult.

We're presenting you with a step-by-step guide to build basic robots, without going into those tough mathematical and technical details. So no matter what your background is, this book is self-sufficient to guide you in making your first robot.

We will start by understanding the basics of robot designing. We will build a fully autonomous robot which will follow a line on its own. There are individual chapters fully dedicated to each aspect of building an autonomous robot; which are mechanics, electronics and programming. Each chapter proceeds in a very lucid way.

Throughout the process of building this robot, we will be working with 12 V DC, so you need not worry about any electrical hazards. The best way to learn something is dive in! We suggest you to actually build a robot along, as you read. Also, don't miss the associated video in this month's DVD.

Chapter 1 begins with an Introduction to robotics. This is aimed at getting you familiar with the different kinds of robots for a better understanding of this exciting field.

Chapter 2 talks about electronics. This is meant for those of you